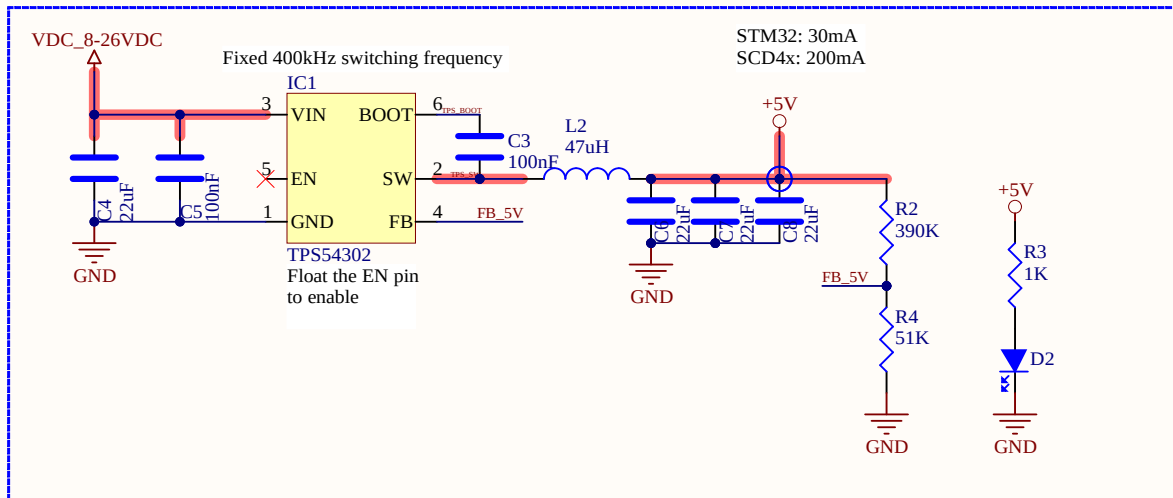
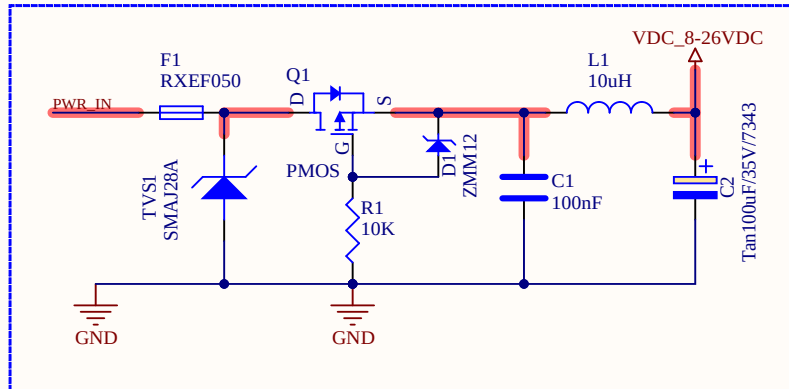
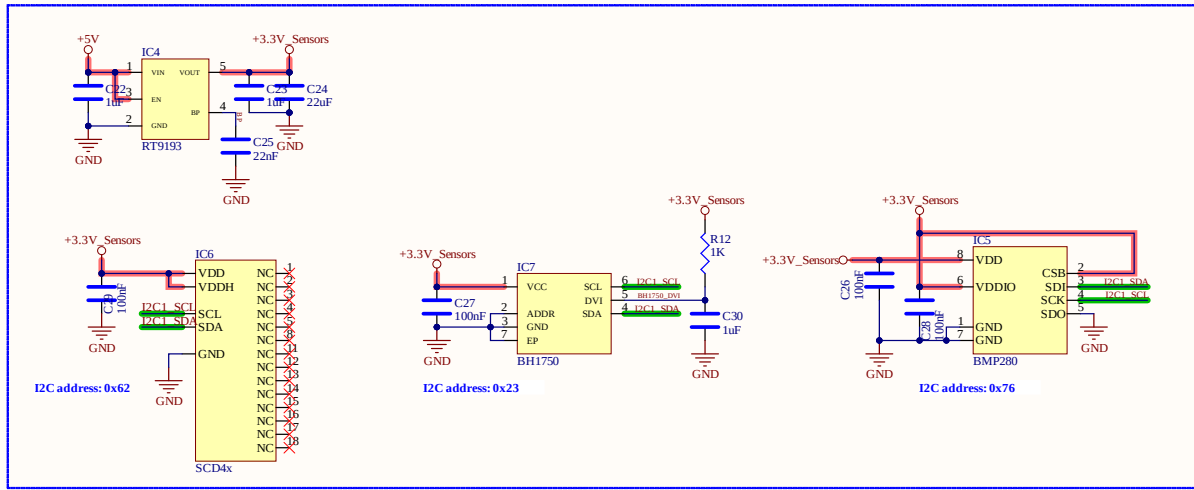
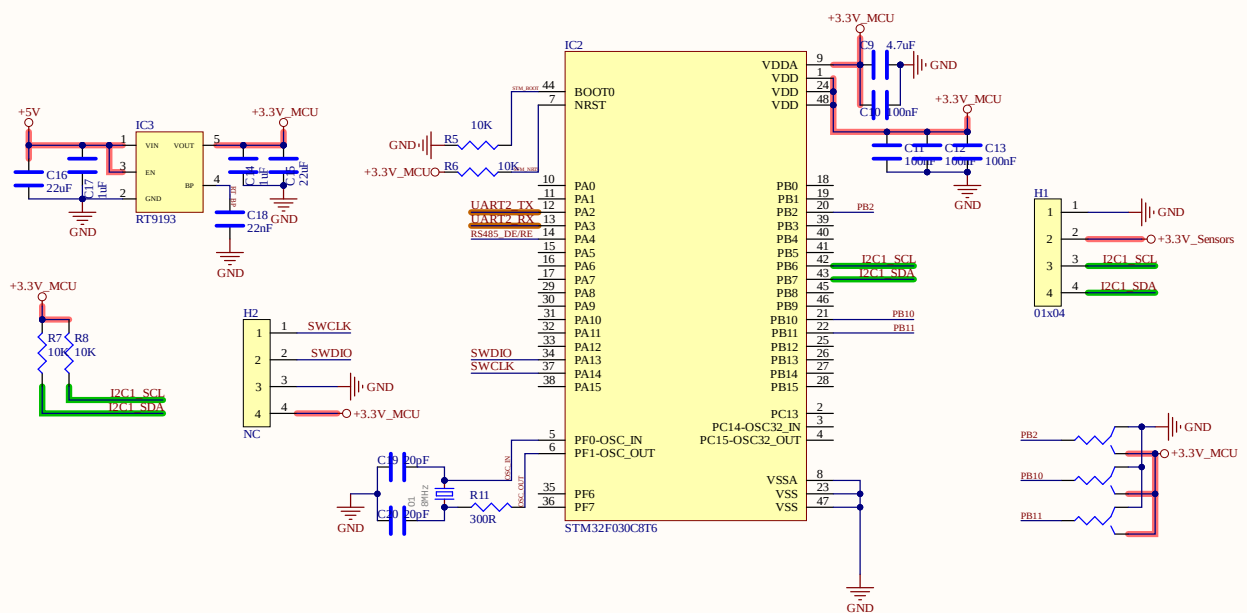
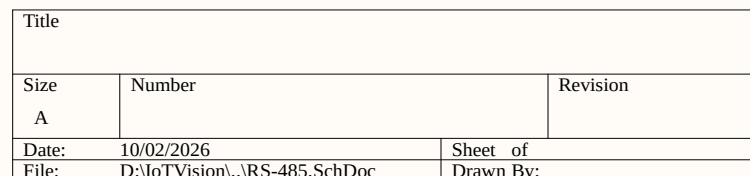


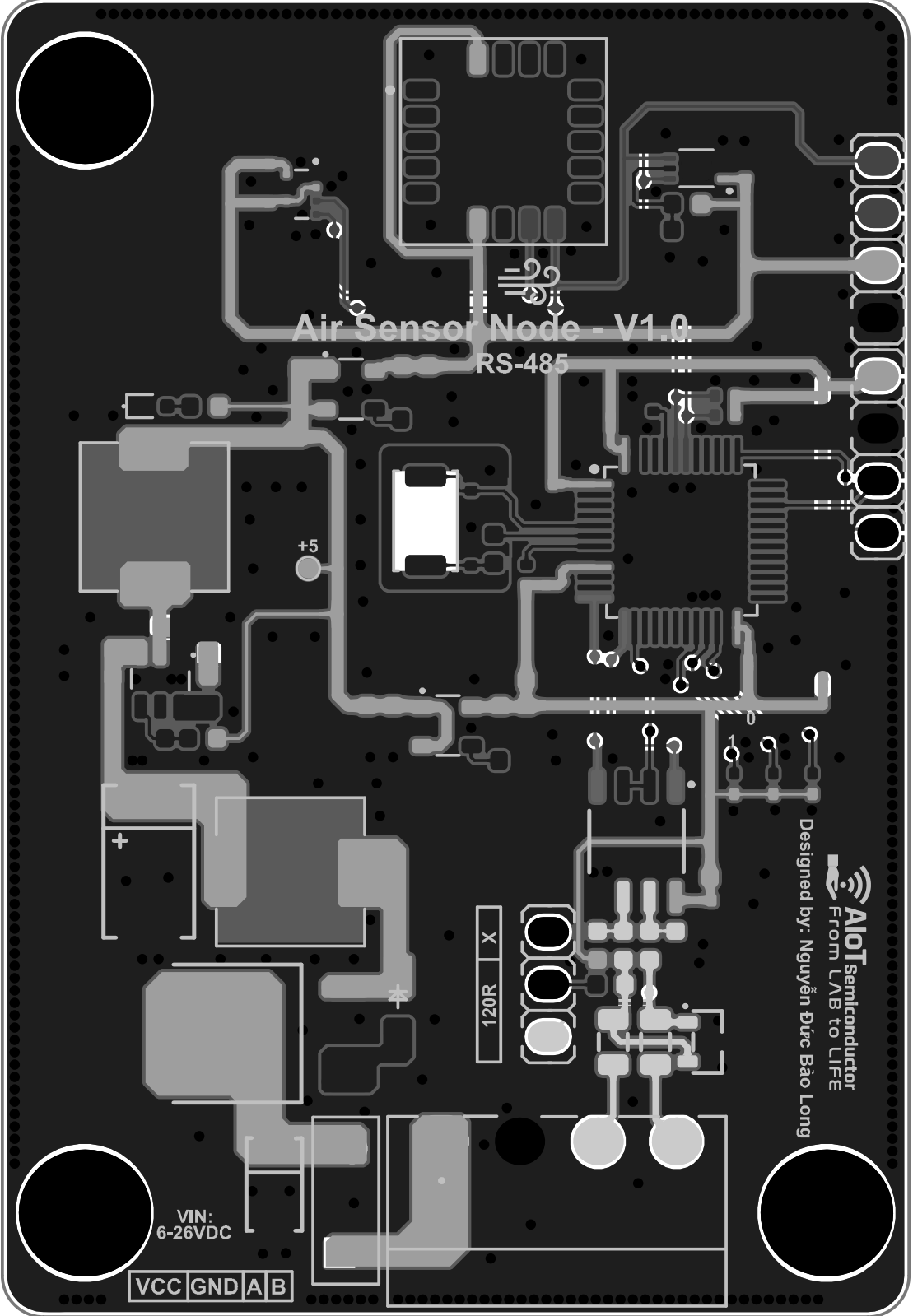


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Title		
Size	Number	Revision
A3		
Date:	10/02/2026	Sheet of
File:	D:\IoT\Vision\...\MCU.SchDoc	Drawn By:





Air Sensor Node - V1.0

RS-485

+5

+

VIN:
6-26VDC

VCC GND A B

120R X

 **AIOT Semiconductor**
From LAB to LIFE
Designed by: Nguyễn Đức Bảo Long

Comment	Description	Designator	Footprint	LibRef	Quantity
100nF	C	C1, C3, C5	C_0603	C	3
Tan100uF/35V/7343	Cp	C2	CP_TANTALUM_EIA_D_7343	Cp	1
22uF	C	C4, C6, C7, C8, C15, C16, C24	C_0603	C	7
4.7uF	C	C9	C_0603	C	1
100nF	C	C10, C11, C12, C13, C26, C27, C28, C29, C32	C_0402	C	9
1uF	C	C14, C17, C22, C23, C30	C_0603	C	5
22nF	C	C18, C25	C_0603	C	2
20pF	C	C19, C20	C_0402	C	2
10uF	C	C31	C_0603	C	1
ZMM12	Diode Zener	D1	SOD-80C_Glass MiniMELF	D_Zener	1
RedLed	LED	D2	LED_0603	LED	1
RXEF050	Fuse	F1	FUSE_0805_2.2x1.5mm	Fuse	1
Fuse	Fuse	F2, F3	FUSE_0805_2.2x1.5mm	Fuse	2
01x04		H1	MALEPIN_1x4	MalePin_1x4	1
NC		H2	MALEPIN_1x4	MalePin_1x4	1
MalePin_1x3		H3	MALEPIN_1x3	MalePin_1x3	1
Hole Via		H4, H5, H6	MountingHole_3.2mm_M3_Pad_Via	Hole	3
TPS54302	Buck Switching Regulator IC,Output 3A SOT-23-6	IC1	SOT-23-6	TPS54302	1
STM32F030C8T6	STM32F030C8T6	IC2	LQFP48	STM32F030C8T6	1
RT9193	RT9193	IC3, IC4	SOT-23-5	RT9193	2
BMP280	Pressure Sensor 4.35PSI ~ 15.95PSI (30kPa ~ 110kPa) Absolute 16b 8-SMD	IC5	BMP280	BMP280	1
SCD4x	SCD4x	IC6	SCD4x	SCD4x	1
BH1750	SENSOR OPT 560NM AMBIENT 6WSOF	IC7	BH1750	BH1750	1
SP3485	Half-Duplex RS-485 Transceiver	IC8	SOIC8_NoPowerPad	SP3485	1
SM712	The SM712 TVS Diode Array is designed to protect RS-485 applications with asymmetrical working voltages (-7V to 12V) from damage due to electrostatic discharge (ESD), electrical fast transients (EFT), and lightning induced surges.	IC9	SOT23-3	SM712	1
Conn_01x04	Conn_01x04	J1	TERMBLOCK_4P_3.81mm_Horizontal	Conn_01x04	1
10uH	Inductor	L1	L_0630_7x7mm	L	1
47uH	Inductor	L2	L_0630_7x7mm	L	1
8MHz		O1	CRYSTAL_5032	CRYSTAL_5032	1
PMOS	MOSFET P	Q1	TO-252-2	MOSFET_P	1
10K	R	R1, R13	R_0603	R	2
390K	R	R2	R_0603	R	1
1K	R	R3, R12	R_0603	R	2
51K	R	R4	R_0603	R	1
10K	R	R5, R6, R7, R8	R_0402	R	4
R	R	R10, R18, R19	R_0402_3Pad	R_3Pad	3
300R	R	R11	R_0603	R	1
120R	R	R14, R16	R_0603	R	2
120R/0603	R	R15	R_0603	R	1
10K/0603	R	R17	R_0603	R	1
TestPoint_1MM		TP2	TestPoint_1MM	TestPoint_1MM	1
SMAJ28A	TVS_D	TVS1	D_SMA	TVS_D_Unidirectional	1